

# Alejandro Zumbado Vargas

## Full Stack Developer

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## SUMMARY

Computer Engineer (TEC, 2019) with 4+ years building end-to-end web products: React/Next.js frontends, Django/Node.js backends, REST APIs, and LLM integrations. Currently at Opal Interactive, building projects for various clients (XR simulators, kiosk/AR apps, AI avatars). In parallel, I design and ship my own products — an AI-driven autonomous content pipeline (newspulse.dev) and published multiplayer WebGL games — owning design, development, deployment (CI/CD), and maintenance.

## SKILLS

**Languages**
JavaScript · TypeScript · Python · C#

**Frontend**
React · Next.js · Astro · Three.js / WebGL · HTML · CSS

**Backend**
Node.js · Express · NestJS · Django · REST APIs

**AI / Data**
LLM integration (Gemini) · RAG pipelines · content automation

**DevOps**
Git/GitHub · GitHub Actions (CI/CD) · Vercel · Firebase · pnpm workspaces

## WORK EXPERIENCE

### Full-Stack Developer — Opal Interactive
Apr 2026 — Present

Studio that takes on projects from various companies and distributes them across its dev team

- Opal Web Avatar: interactive AI avatar platform — React 19, Express, lipsync, RAG.
- HealthSim: Unity XR training simulator for medical equipment, with remote sessions via Netcode + Unity Relay.
- totempanadol: Unity kiosk/AR app with thermal camera, face detection, and Firebase.
- Contributed to Startus: a 4-repo SaaS platform (NestJS, React, FastAPI, Chrome extension).

### React / Frontend Developer — Freelance — independent projects
2021 — Present

Multiple clients and projects

- Solucom — Remote Gate Access (2023): IoT access-control system with a Raspberry Pi (Python) as the physical controller, Vue.js frontend, and Laravel/PHP backend, synced over an API with a password database.
- Built applications and components with React: clean architecture, reusable components, responsive design.
- Integrated third-party APIs, fixed bugs, and improved performance under tight deadlines.

### Full-Stack Developer (Jr/Mid) — IMMERSO
Apr 2021 — Jan 2023

- Built and maintained frontends with React.js and Next.js connected to Django backends.
- Built interactive web and VR experiences with A-Frame.js: editors, maps, and database-driven content.

**Full-Stack Developer (Jr/Mid) — WOW Emotions** Jan 2022 — Feb 2023

- Led end-to-end development of a VR-enabled website for the 'Doit' reconstruction project.
- Designed and built 'Giz', an immersive nature experience with 5+ VR levels.
- Optimized performance and visual consistency across devices and browsers.

**Full-Stack Developer (Jr) — SolvoSoft** Aug 2020 — Feb 2021

- Developed and optimized web features with JavaScript and Django on legacy systems.
- Collaborated with cross-functional teams to improve UI performance and responsiveness.

## INDEPENDENT PROJECTS

**NewsPulse — personal product — newspulse.dev** Jun 2026 — Present

Fully autonomous, AI-generated news pipeline

- Python pipeline that researches news via RSS, writes articles in 4 languages with an LLM, and fact-checks them against source material before publishing.
- Static frontend built with Astro + Tailwind, deployed on Vercel; publishing runs automatically on a GitHub Actions cron with no human in the loop.

**Tinted Showdown — personal game — published** 2025 — Present

2–4 player multiplayer, PC / Android / WebGL

- Multiplayer Unity game with Netcode for GameObjects + Unity Relay; live, playable WebGL build.
- Extracting the networking layer as a standalone package (QuickLobby — Netcode Relay Kit), in development for release as a Unity asset.

**Cannons — personal game — published** Aug 2025 — Present

Tower-defense, 500 levels

- Unity tower-defense with progression across up to 500 levels and a self-play Python AI that plays and designs new levels.

**FeedBack Figuras 3D — personal product** 2025 — Present

Customer feedback collection system

- Next.js 15 + React 19 app for rating custom 3D-printed figures by category, uploading photos, and exporting results from an admin panel.
- Backend with Supabase, validation with Zod and React Hook Form, animations with Framer Motion.

## EDUCATION

**Computer Engineering** 2019

Tecnológico de Costa Rica (TEC)

**High School Diploma** 2015

Liceo San Rafael de Alajuela

## LANGUAGES

Spanish — Native

English — Intermediate